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Report

Feasibility of a new 10-100 MW plant

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Wind farm project – Alberta, Canada

Canada has significant renewable energy potential due to its abundant wind resources, particularly in regions like Alberta, Ontario, and Quebec.[1] This report evaluates the feasibility of a 60 MW wind farm in southern Alberta, using RETScreen projections for guidance on production and costs.

The wind farm project will be located in the province of Calgary, in southern Alberta near the border. This region has favourable wind conditions, with significant wind energy potential.[2] While Alberta has historically been dominated by fossil fuels, it is undergoing a transition towards renewable energy. As shown in figure 1, the province's emissions totalled 269.9 megatons of CO2 equivalent in 2022 reflecting a 52% increase since 1990. [3]

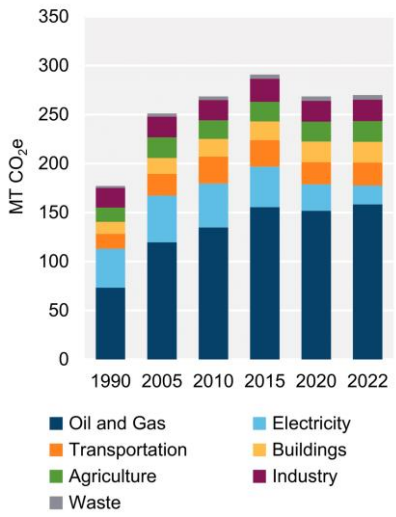

Figure 1: GHG Emissions by Sector in Alberta, Canada[3]

Therefore, selecting southern Alberta addresses the need for energy diversification by developing a wind project in a region less saturated than Ontario or Quebec. Moreover, land costs in Alberta are generally cheaper than those in Ontario or other provinces, and available space allows for project development with fewer constraints related to urbanization.

In fact, the project will consist of modern turbines with a capacity of 2,000 kW each, sourced from suppliers such as AAER or GE Energy. The turbines will have heights ranging from 70 to 100 meters. The farm's total installed capacity will be 60 MW (60,000 kW), with a capacity factor of 28.5%. This factor indicates that the farm is expected to operate at this efficiency level, producing approximately 150 025 142 864 kWh of electricity annually.[4]

Moreover, the 60 MW wind farm project requires an initial investment of CAD 126 million for the installation of turbines and associated infrastructure according to RETScreen. This financing is structured with debt representing 70% of the total amount (CAD 88.2 million) and an equity contribution of CAD 37.8 million. Debt repayment is planned over 15 years at an interest rate of 7%, resulting in annual payments of CAD 9.68 million. Operating and maintenance costs are estimated at CAD 2.04 million per year. The projected annual electricity production is approximately 150 GWh, sold at CAD 0.10 per kWh, generating annual revenues of CAD 15 million. Consequently, the project achieves a positive net cash flow of CAD 3.26 million from the first year. The Net Present Value over 20 years is projected at CAD 7.76 million, with a return on investment expected in 10.9 years.[4]

Besides, Southern Alberta is attractive to wind projects due to favourable wind resources. Projects such as Castle Rock Ridge and others are already well established in the area.[5] Compared to these larger-scale projects, the 60 MW wind farm project positions itself as a medium-sized project that effectively meets local energy needs and integrates well into the existing grid.

Last but not least, the Canadian government offers various subsidy programs aimed at promoting renewable energy, including the Smart Renewables and Electrification Pathways Program.[6] These programs could provide funding for part of the initial investment costs, thereby reducing the Capital Expenditures and improving the project's net cash flow.

In conclusion, the 60 MW wind farm project in southern Alberta is economically feasible, with installation costs of CAD 126 million and projected annual revenues of CAD 15 million ensuring positive cash flow from year one. Available subsidies enhance its financial viability, positioning it as a key initiative to diversify Alberta's energy supply and support greenhouse gas reduction efforts.

References

[1] "Wind Energy." Natural Resources Canada. https://natural-resources.canada.ca/energy/energy-sources-distribution/renewables/wind-energy/7299 (accessed Oct. 5, 2024).

[2] "Wind energy development." Alberta Wilderness Association. https://albertawilderness.ca/issues/wildlands/energy/wind-power/ (accessed Oct. 6, 2024).

[3] "Provincial and Territorial Energy Profiles – Alberta." Canada Energy Regulator. https://www.cer-rec.gc.ca/en/data-analysis/energy-markets/provincial-territorial-energy-profiles/provincial-territorial-energy-profiles-alberta.html#s2 (accessed Oct. 6, 2024).

[4] "RETScreen." RETScreen Clean Energy Management Software. Natural Resources Canada. https://natural-resources.canada.ca/maps-tools-and-publications/tools/modelling-tools/retscreen/7465 (accessed Oct. 5, 2024).

[5] "Alberta now has 44 wind farms, and Wednesday night they collectively produced next to no power." Pipeline Online. https://pipelineonline.ca/alberta-now-has-44-wind-farms-and-wednesday-night-they-collectively-produced-next-to-no-power/ (accessed Oct. 6, 2024).

[6] "Smart Renewables and Electrification Pathways Program." Natural Resources Canada. https://natural-resources.canada.ca/climate-change/green-infrastructure-programs/sreps/23566 (accessed Oct. 6, 2024).